# 3 (1)

## Classification and tabulation of Statistical data

## Classification and tabulation of statistical data

The collected data, also identified as raw data or ungrouped data is always in an unorganized form and requires to be organized and displayed in a ***meaningful*** and ***readily understandable*** form in order to help further statistical analysis.

# Classification and tabulation of statistical data

***Definition*:-** Classification is the process of arranging data into homogeneous (similar) groups according to their common characteristics.

## Classification and tabulation of statistical data

For example, the population of a town can be grouped according to sex, age, marital status, etc.

## Classification and tabulation of statistical data(Contd. . . )

### *Types of classifications:*

- Spatial /Geographical classification
- Chronological classification
- Qualitative classification
- Quantitative classification

## **Classification and tabulation of statistical data** (Contd. . . )

### ***Geographical / Spatial classification***

Under this type of classification, the data are classified on the basis of area or place, and as such, this type of classification is also known as ***areal or spatial classification.***

The areas may be in terms of countries, states, districts, or zones according as the data are distributed.

## **Classification and tabulation of statistical data**(Contd. . . )

This type of classification is suitable for those data which are distributed geographical relating to a phenomenon namely population, mineral resources, production, sales, students of universities etc.

***Example1*** :- Coffee suppliers of the different regions of Ethiopia to the central government.

***Example2*** :- The following table refers list of countries by coffee production.

# Classification and tabulation of statistical data(Contd. . . )

| Rank | Country | Metric tones |
|---|---|---|
| 1 | Brazil | 2,652,000 |
| 2 | Vietnam | 1,650,000 |
| 3 | Colombia | 810,000 |
| 4 | Indonesia | 660,000 |
| 5 | Honduras | 580,000 |
| 6 | Ethiopia | 384,000 |

## *Chronological classification*

Under this type of classification, the data collected are classified on the basis of time of their occurrence.

As such, the series obtained under this classification is purely known as a time series. This type of classification is suitable for chose data which take place in course of time namely population, production, sales, results etc.

## Classification and tabulation of statistical data(Contd. . . )

The different classes obtained under this classification are arranged in order of the time which may begin either with the earliest, or the latest period.

***Example3*** :- The following table refers to the second Growth & Transformation Plan of the government of Ethiopia (*Ethiopia Coffee Annual Report: Global Agriculture Information Network*.)

| Table 1: Status of GPT II Production Target (1,000 Metric Tons) | | | | | |
|---|---|---|---|---|---|
| | **2015/16** | **2016/17** | **2017/18** | **2018/19** | **2019/20** |
| **GTP II Target** | **504** | **605** | **726** | **871** | **1,103** |
| **Production** | **391** | **417** | **423** | **438** | **444** |
| **% Achieved** | **78** | **69** | **58** | **49** | **40** |

## *Qualitative Classification*

Under this type of classification, the data obtained are classified based on certain descriptive character or qualitative aspect of a phenomenon namely sex, citizenship, qualification, marital status, religion etc.

## *Quantitative Classification*

Under this type of classification, the collected data are classified on the basis of certain variable namely mark, income, expenditure, profit, loss, height, weight, age, price, production etc

## **Classification and tabulation of statistical data**(Contd. . . )

We divide quantitative classification in to two variables:

- *Discrete variables*
- *Continuous variables*

## **Classification and tabulation of statistical data**(Contd. . . )

*Discrete* and *continuous* variables are two broad categories of numerical data. Numeric variables represent characteristics that you can express as numbers rather than descriptive language.

***Discrete variables*** can only assume specific values that you cannot subdivide. Typically, you count discrete values, and the results are integers.

# Classification and tabulation of statistical data(Contd. . . )

For example, if you work at a zoo, you'll count the number of lions.

Discrete data can only take on specific values. For example, you might count 20 lions at the zoo.

These variables cannot have fractional or decimal values. You can have 20 or 21 lions, but not 20.5!

Natural numbers have discrete values.

Other examples of discrete variables include the following:

## Classification and tabulation of statistical data(Contd. . . )

- The number of books in the library.
- The number of heads in a sequence of coin tosses.
- The result of rolling a die.
- The number of patients in a hospital.

## Classification and tabulation of statistical data(Contd. . . )

While discrete variables have no decimal places, the average of these values can be fractional.

## Classification and tabulation of statistical data(Contd. . . )

For example, families can have only a discrete number of children:

1, 2, 3, etc. However, the average number of children per family can be 2.2.

## Classification and tabulation of statistical data(Contd. . . )

***Continuous variables*** can assume any numeric value and can be meaningfully split into smaller parts. Consequently, they have valid fractional and decimal values.

In fact, continuous variables have an infinite number of potential values between any two points. Generally, you measure them using a scale.

## Classification and tabulation of statistical data(Contd. . . )

When you see decimal places for individual data points, you’re looking at a continuous data.

## **Classification and tabulation of statistical data**(Contd. . . )

Examples of continuous variables include *weight, height, length, time, temperature, salary of workers, marks of students, telephone bills* etc.

## Frequency Distributions

Frequency Distributions are tabular representations of a quantitatively classified data.

## **Classification and tabulation of statistical data**(Contd. . . )

We divide Frequency Distributions into two measure groups:

- *Ungrouped Frequency Distributions*
- *Grouped Frequency Distributions*

## **Classification and tabulation of statistical data**(Contd. . . )

Ungrouped data is data given as individual data points. Grouped data is data given in intervals.

## Classification and tabulation of statistical data (Contd. . . )

***Example1:-*** The following table refers to the marks in statistics and probability course (out of 25) of the sample of 20 students at HiLCoE in Spring 2025 term.

## Classification and tabulation of statistical data(Contd

<table>
<tr><th>Marks Obtained</th><th>Tally Marks</th><th>Frequency</th></tr>
<tr><td>12</td><td>|</td><td>1</td></tr>
<tr><td>15</td><td>||</td><td>2</td></tr>
<tr><td>17</td><td>|||</td><td>3</td></tr>
<tr><td>19</td><td>~~||||~~</td><td>5</td></tr>
<tr><td>21</td><td>|||</td><td>3</td></tr>
<tr><td>23</td><td>||||</td><td>4</td></tr>
<tr><td>25</td><td>||</td><td>2</td></tr>
</table>

## *Grouped Frequency Distribution*

***Definition:-*** A Grouped Frequency Distribution (GFD) is a table in which the values for a variable are grouped into classes together with the numbers of observed values falling into each class.

The number of observed values that belong to class is called its ***frequency***.

Below listed are terms we need to know to construct a Grouped Frequency Distribution.

## *Grouped Frequency Distribution*

- Number of Classes
- Class Width
- Class Limits
- Class Boundaries
- Class Mark

# Grouped Frequency Distribution

- Frequency
- “Rules” for forming a GFD
- Graphs of a GFD
  - Histogram
  - Frequency Polygon

# Grouped Frequency Distribution

***Example 2:-*** The following table refers to the weight (to the nearest kg) of a sample of 50 students of the 2025 graduating class of the computer science department of Jimma University.

# Grouped Frequency Distribution

| Weight (in kg) | Number of students |
| --- | --- |
| 45 – 52 | 5 |
| 53 -60 | 8 |
| 61 – 68 | 13 |
| 69 – 76 | 16 |
| 77 – 84 | 5 |
| 85 - 92 | 3 |

# Class Limits, Class Boundaries and Frequency

| Class Limit | Class Boundary | Frequency |
|---|---|---|
| 45 – 52 | 44.5 – 52.5 | 5 |
| 53 -60 | 52.5 - 60.5 | 8 |
| 61 – 68 | 60.5 – 68.5 | 13 |
| 69 – 76 | 68.5 – 76.5 | 16 |
| 77 – 84 | 76.5 – 84.5 | 5 |
| 85 - 92 | 84.5 – 92.5 | 3 |

# Class Limits, Class Boundaries, Class Marks and Frequency

| Class Limit | Class Boundary | Class Mark | Frequency |
|---|---|---|---|
| 45 – 52 | 44.5 – 52.5 | 48.5 | 5 |
| 53 -60 | 52.5 - 60.5 | 56.5 | 8 |
| 61 – 68 | 60.5 – 68.5 | 64.5 | 13 |
| 69 – 76 | 68.5 – 76.5 | 72.5 | 16 |
| 77 – 84 | 76.5 – 84.5 | 80.5 | 5 |
| 85 - 92 | 84.5 – 92.5 | 88.5 | 3 |

# *"Rules" for forming a GFD*

## ***Rule #1.*** **Choose the classes**

Using less than 5 or More than 15 classes is not recommended. The reason is to make your table clear and easily understandable.

## **Grouped Frequency Distribution**(Contd. . . )

You will normally be told how many classes you need.

***Rule #2.*** ***Choose the class width*****.**

The ***class width*** is the range of the class.

**Always round up the class width to the accuracy of the given data. This means:**

## **Grouped Frequency Distribution**(Contd. . . )

- if your data do not have decimals at all, then round up the class width to the nearest unit.
- if your data have numbers given with one decimal place, then round up the class width to the nearest tenths.

## Grouped Frequency Distribution(Contd. . . )

- As much as possible, try to have equal width to each class. This avoids the distorted view of data.

***FINDING CLASS WIDTH***

- The class width can be found by taking the ratio of the difference of the smallest data from the largest data by the number of classes.

## Grouped Frequency Distribution(Contd. . . )

i.e. Class width =

$$\frac{x_{Max} - x_{Min}}{Number\ of\ Classes}$$

## ***Rule #3.*** **Forming Class Limits**

**Class Limit:** The values which determine the upper and lower limits of a class.

## Grouped Frequency Distribution(Contd. . . )

**Lower Class Limit:** Smallest data value that can be included in the class.

**Upper Class Limit:** Largest data value that can be included in the class.

**Class boundaries**: They are halfway points that separate the **classes**.

## **Grouped Frequency Distribution**(Contd. . . )

**Lower class boundary** of a given **class** is obtained by averaging the upper limit of the previous **class** and the lower limit of the given **class**.

## **Grouped Frequency Distribution**(Contd. . . )

**Upper class boundary** of a given **class** is obtained by averaging the upper limit of the previous **class** and the lower limit of the given **class**.

## **Grouped Frequency Distribution**(Contd. . . )

*Class Limits are Mutually Exclusive*.

This means class limits cannot overlap or be contained in more than one class.

You have to note that class boundaries are not ***Mutually Exclusive***.

## **Grouped Frequency Distribution**(Contd. . . )

And this means the upper class boundary of the first class serves as the lower class boundary of the second class. The upper-class boundary of the third class serves as the lower-class boundary of the fifth class and so on.

# Grouped Frequency Distribution(Contd. . . )

## *CALCULATING A CLASS MARK (ALSO KNOWN AS THE CLASS MID POINT)*

***The class mark is defined to be the average of the class interval. That is***

$$\text{Class Mark} = \frac{\text{Upper Class Limit} + \text{Lower Class Limit}}{2}$$

$$\text{Class Mark} = \frac{\text{Upper Class Boundary} + \text{Lower Class Boundary}}{2}$$

## **Grouped Frequency Distribution**(Contd. . . )

***Rule #4.*** ***Exhaustiveness***

There should be enough classes to accommodate all of the data.

***Example1*** :- The following table refers to the weight of the sample of 77 students to the nearest KG.

| ~~Class Limit~~<br>Wt. of students<br>(in KG) | ~~Frequency~~<br>Number of Students | Class Boundary | Class Mark |
|---|---|---|---|
| 35 – 43 | 8 | 34.5 – 43.5 | 39 |
| 44 – 52 | 11 | 43.5 – 52.5 | 48 |
| 53 – 61 | 16 | 52.5 – 61.5 | 57 |
| 62 – 70 | 19 | 61.5 – 70.5 | 66 |
| 71 – 79 | 14 | 70.5 – 79.5 | 75 |
| 80 - 88 | 9 | 79.5 – 88.5 | 84 |

***Example2*** :- The following data refers to the marks in Statistics and Probability of Spring 2025 term (out of 100% to the nearest unit) of the sample of HiLCoE students.

***(i)*** Construct a GFD using **6 classes**

***(ii)*** Since the data given is a continuous data, your GFD should consist of Class Limit, Class Boundary, Class Mark and Frequency.

## Grouped Frequency Distribution(Contd. . . )

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 83 | 29 | 40 | 48 | 55 | 59 | 61 | 66 | 73 |
| 75 | 61 | 42 | 85 | 59 | 46 | 32 | 88 | 79 |
| 61 | 54 | 65 | 63 | 26 | 85 | 38 | 77 | 49 |
| 68 | 75 | 60 | 41 | 62 | 90 | 38 | 54 | 61 |
| 68 | 71 | 58 | 64 | 77 | 50 | 56 | 63 | 73 |
| 33 | 46 | 51 | 71 | 54 | 64 | 65 | 63 | 75 |

# *Any Questions?*


Thank
you